



Education

University of Waterloo

BASc Biomedical Engineering – 86.31% Average

Sep 2021 - Apr 2026

Waterloo, ON

Experience

Electrical Engineer

OtO Inc.

Jan - Aug 2025

Toronto, ON

- Designed **electrical schematics** and **PCBs** in **Altium**; designed PCB **bring-up** testing suite for hardware debugging/validation
- Built prototype PCBs using OpenPNP; validated performance using **oscilloscopes**, **spectral analyzers**, **logic analyzers**, DMMs
- Supported **EMI/EMC** design and measurement for prototype PCBs, participated in verification at EMI compliance test facilities
- Developed firmware features for **ESP32**; temperature-controlled charging, **I2C** motor controllers, integrating new ICs, bugfixes
- Reduced device sleep current by **34%**, boosted charging speed by **47%**, and implemented new battery-failure operation features
- Implemented **CI workflows** with hardware emulation, unit tests, and static code analysis for improved code robustness

Hardware and Software Developer

Elev8 & Perform

May - Aug 2024

Toronto, ON

- Spearheaded **prototype** device development; BLE for data transmission, I2C for gas/flow sensors, handling tasks with FreeRTOS
- Designed PCB integrating ESP32 and sensors; **modeled and 3D printed** device housing for ergonomics and functionality
- Implemented Python-based **data visualization** solutions for comprehensive sensor data analysis, testing, and system validation
- Developed Python Flask API and web app on Google Cloud, visualizing live device data, and saving to MongoDB collection

Firmware Developer

Onsemi

Sep - Dec 2023

Waterloo, ON

- Developed embedded software for RSL15 MCU (**ARM Cortex-M33**), SDK sample applications, and **fixing customer bugs**
- Implemented **GPIO drivers** on HAL libraries and components, and customer-facing SDK firmware in C (**AUTOSAR**)
- Implemented **unit testing** (CppUTest); hardware testing with DMM, logic analyzers, STM32CubeMonPwr, and JTAG debuggers

Undergraduate Research Assistant

University of Waterloo - Department of Nanotechnology Engineering

Jan 2024 - Jan 2025

Waterloo, ON

- Synthesized electroretinography (ERG) electrodes with Au-Ca and Au-Pt nanoparticles for **R&D of diagnostic sensor strips**
- Validated in-lab electrodes using oscilloscope, DMM; analyzed signal data with Excel and Python, **10% improved performance**

Junior Systems Engineer

Ontario Health

Jan - Apr 2023

Toronto, ON

- Developed serverless Electronic Medical Record (EMR) web app on Azure using AngularJS, JavaScript, Node.js, and MongoDB
- Optimized RESTful API endpoints and backend, resulting in **94% development cost reduction** and improved EMR accessibility

Skills

- **Software:** C, C++, Python, Rust, MATLAB, FreeRTOS, CMake, CppUTest, ESP-IDF
- **Hardware:** PCB Design/Bring-Up, Soldering, Oscilloscope, Logic Analyzer, Spectral Analyzer, DMM, ESP32, RPi, Jetson
- **Tools:** Git, Linux, Altium, KiCad, LTSPICE, Coverity, Jama, VSCode, Eclipse, SolidWorks, Jira

Projects

Electrical Lead

True North Biomedical Competition - Electrical Team

Sep 2025 - Apr 2026

Waterloo, ON

- Led a 9-member team, spearheaded electrical development for knee-brace device, designing electrical schematics, PCB design

Embedded/Electrical Lead

VocalPoint - Capstone Project

- Led electrical development and firmware architecture for a table-top microphone array Bluetooth device for hearing-aids
- Designed electrical schematic and PCB in Altium for a RPi CM5 carrier board; USB-C power, I2S multiplexer, PCIe connectors
- Developed BLE and I2C applications for the ESP32 and RPi CM5 for telemetry communication